Deep Learning Based Speech Interview Analysis With Multiagent Feedback

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***Abstract*—This project presents an audio-only deep learning-based voice interview coach for predicting interview-related performance scores and generating understandable feedback. Using the RecruitView dataset, interview videos were converted into audio files. The study evaluates three speech-based models: an acoustic feature model, a HuBERT/wav2vec2 embedding model, and a new HuBERT/wav2vec2 model that combines acoustic features. The models predicted speaking skills, confidence score, interview score, and overall performance. Results showed that the HuBERT embedding model achieved the best overall performance, with an average MAE of 0.7082 and Pearson correlation of 0.4138. The fusion model performed better than the acoustic baseline but did not outperform HuBERT alone. Finally, predicted scores and acoustic indicators were passed to a multi-agent feedback layer to generate human-readable interview coaching feedback. This project provides a lightweight and privacy-conscious audio-only option compared to multimodal interview assessment systems.**

I. Introduction and Workload Distribution

Job interviews are an important part of the recruitment process, and candidates are often judged not only by what they say, but also by how they communicate. Speech delivery, confidence, fluency, pauses, tone, and energy can influence the impression created during an interview. Many candidates struggle with nervousness, long pauses, low energy, unclear speech, or unstructured answers. Therefore, an automated interview coaching system can be useful for helping candidates practice and improve their communication skills.

Earlier automated interview analysis research has shown that behavioral signals can be useful for predicting interview-related traits. Naim et al. proposed a computational framework for analyzing job interview performance using facial expressions, language features, and prosodic speech features such as pitch, intonation, and pauses [1]. Their work showed that interview-related traits such as engagement and friendliness can be predicted using automatically extracted behavioral cues.

Later work such as HireNet moved toward large-scale deep learning-based interview analysis. HireNet used asynchronous video interview data from more than 7000 candidates and proposed a hierarchical attention model for hirability prediction [2]. More recently, RecruitView introduced a multimodal dataset with 2,011 naturalistic video interview clips from more than 300 participants and 27,000 pairwise comparative judgments across personality and interview-performance dimensions. Along with the dataset, the authors proposed CRMF, a geometric deep learning framework that fuses audio, video, and text representations using hyperbolic, spherical, and Euclidean manifolds [3].

Although these systems show strong performance, they mainly focus on multimodal interview assessment or hirability prediction. In contrast, this project focuses on an audio-only interview coaching system. The goal is to analyze only the candidate's voice and generate useful self-improvement feedback for interview preparation. The project is not designed to make hiring or rejection decisions.

The feedback is generated using a multi-agent feedback layer. After the best-performing deep learning model predicts interview-related scores and acoustic indicators, these outputs are passed to four agents: a Judge Agent, Critic Agent, Supporter Agent, and Coach Agent. The Judge Agent gives an overall evaluation, the Critic Agent identifies weaknesses, the Supporter Agent highlights strengths, and the Coach Agent combines all observations into final actionable feedback [4]. In this way, the multi-agent layer converts numerical model outputs into understandable coaching advice for the candidate.

The deep learning model is the central component of this project. The multi-agent system is used only as a final explanation layer that converts model outputs into structured coaching feedback.

TABLE I
Workload Distribution

Task	Responsible Member(s)
Literature review on automated interview systems and speech emotion recognition	All members
RecruitView dataset understanding and preprocessing	All members
Audio extraction from video clips	Saketh Yadav
Acoustic feature extraction	Saketh Yadav
Acoustic baseline implementation	Karthika Veeravaneni
HuBERT/wav2vec2 embedding baseline implementation	Karthika veeravaneni
Proposed fusion model implementation	Charvi Sri Bodagala
Model training and evaluation	Charvi Sri Bodagla
Multi-agent feedback design	Charvi Sri Bodagala
Report writing, result analysis, and presentation	All members

II. Problem Statement

A. Problem Statement

Interview candidates often receive only limited feedback on how they deliver their answers. Numerical scores or general comments are not always enough to help a candidate understand specific speaking-related issues such as hesitation,

low confidence, long pauses, fast speaking rate, or weak vocal energy. At the same time, many automated interview-analysis systems rely on multimodal inputs such as video, facial expressions, body language, text, and audio, which can make the system more complex and less privacy-friendly.

The problem addressed in this project is, therefore, to investigate whether interview-related performance scores can be predicted using only the audio signal and whether these predictions can be converted into useful coaching feedback that is easier for the user to understand.

Note: It is intended to support, not to give hiring decisions

III. Problem Definition

A. Motivation

Interview performance depends not only on what a candidate says, but also on how the answer is delivered. A candidate may have good knowledge but may still sound hesitant, nervous, unclear, or low in confidence due to long pauses, low energy, fast or slow speaking rate, or limited vocal variation. These delivery-related factors can affect how an answer is perceived in an interview.

The motivation of this project is to go beyond raw prediction scores. A model may predict that a candidate has low confidence or weak speaking skills, but the user still needs to know what to improve. Therefore, the project includes an agentic feedback layer that converts predicted scores and acoustic indicators into structured coaching feedback. This makes the output more understandable and useful for interview preparation.

B. Aim

This project aims to develop and evaluate an audio-only deep learning-based voice interview coach that predicts interview-related scores from speech and converts the predicted scores and acoustic indicators into structured coaching feedback using a multi-agent feedback layer.

C. Objectives

The objectives of this project are:

1) To prepare the RecruitView dataset for audio-only modelling by extracting WAV audio files from interview video clips.
2) To create user-independent train, validation, and test splits using the participant identifier `user_no`.
3) To extract handcrafted acoustic features from interview audio recordings, including duration, speaking rate, pause ratio, voiced ratio, pitch statistics, and energy statistics.
4) To implement an acoustic feature baseline model using handcrafted speech-delivery features.
5) To implement HuBERT/wav2vec2 embeddings baseline model
6) To implement a fusion model that combines HuBERT embeddings with handcrafted acoustic features.
7) To compare the implemented models using regression metrics.
8) To select the best-performing model as the final prediction layer for the interview coaching system.
9) To use a multi-agent feedback layer consisting of judge, critic, supporter, and coach agents to convert model outputs into understandable interview coaching feedback based on the predictions.

D. Hypotheses

This project is guided by the following hypotheses:

H1: Combining pretrained HuBERT speech embeddings with handcrafted acoustic features will improve interview-score prediction compared with using HuBERT embeddings alone, because acoustic indicators such as speaking rate, pause ratio, pitch variation, and energy may provide additional delivery-related information.

H2: A multi-agent feedback layer can make the system more useful for interview coaching by transforming numerical prediction scores and acoustic indicators into understandable and actionable feedback, instead of presenting raw model outputs alone.

IV. Background and Related Work

A. Automated Job Interview Analysis

People have been studying automated interview analysis using machine learning and deep learning methods. Naim et al. Looked at interview videos from students who wanted internships and extracted features like what they said, their expressions and how they sounded [1]. Their framework tried to predict how the students did in the interview like how excited or friendly they were. It found that things like pitch and pauses in their speech were important.

This work is important because it shows that how people speak can tell us a lot about how they do in interviews. However it used a lot of things like facial expressions and language while this project only looks at audio.

B. HireNet

HireNet [2] is a model that uses attention to look at video job interviews. It looks at the answers the questions and the job the person is applying for to try to figure out if they should be hired. The people who made HireNet collected data from job interviews and treated each interview like a series of questions and answers. Their model did better than approaches that looked at what people said heard or saw in the videos.

HireNet is a method because it uses deep learning to look at real job interviews. However it is trying to figure out if someone should be hired while this project is trying to help people improve their interview skills.

C. RecruitView and CRMF

RecruitView is a dataset that has a lot of video clips from job interviews [3]. It has over 2,000 clips from than 300 people and it tries to figure out their personality and how they do in interviews. The people who made RecruitView also made a model called CRMF. CRMF is a framework that looks at how

people behave in ways like how they speak or move and it tries to figure out their personality and how they do in interviews.

CRMF is very important for this project because it is one of the methods for looking at multiple things at once. The people who made CRMF say that it does better than models that look at multiple things and it uses fewer resources.

However CRMF looks at audio, video and text together. This project only looks at audio. So CRMF is not exactly the same. It is a good model to compare to. The goal of this project is to see if a simpler model that only looks at audio can help people improve their interview skills without needing much information and being easier to use.

D. Speech Emotion Recognition and Pretrained Speech Models

Speech emotion recognition is when a model tries to figure out how someone is feeling based on how they speak. Traditional models use features that people made while newer models use deep learning.

Chen et al. proposed a framework that uses a model called wav2vec 2.0 [5]to recognize emotions in speech. Their model can transcribe speech. Recognize emotions at the same time and it did very well on a benchmark test. Another person proposed a system that can recognize emotions in languages using wav2vec 2.0 and it did better, than other models that only looked at one language.

These studies support the use of models like HuBERT or wav2vec2 to look at how people speak and figure out how they are feeling [6], [7].

E. Research Gap

Existing automated interview-analysis systems have mainly developed in two directions. The first direction focuses on multimodal interview assessment, where video, audio, facial expressions, text, and behavioral cues are combined to predict interview performance or hirability. The second direction focuses on speech emotion recognition, where audio is used to classify general emotion labels such as happy, sad, angry, neutral, or fearful.

Recent systems such as RecruitView and CRMF show strong progress in multimodal interview assessment. However, they rely on multiple modalities, including video, audio, and text. This can make the system computationally heavier and may also raise privacy concerns because facial and visual behavioral data are involved. In contrast, many interview preparation scenarios may only require voice-based feedback, where the candidate receives guidance on speech delivery without sharing video data.

Therefore, the main gap addressed in this project is the lack of a lightweight audio-only interview coaching system that combines deep speech representations, acoustic delivery features, interview-related score prediction, and structured feedback generation.

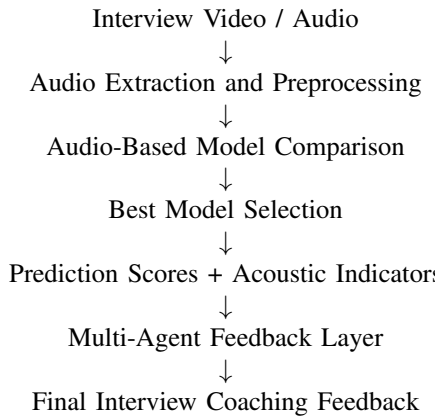


Fig. 1. Overall architecture of the proposed voice interview coach.

V. Methodology / Proposed Approach

A. Overall Architecture

The proposed system follows the pipeline refer the figure:

The deep learning prediction model is the main component. The agent layer is secondary and only transforms model outputs into feedback.

B. Dataset

The selected dataset is RecruitView. It contains multimodal interview clips and interview-related labels. For this project, only the audio part of the videos is used.

RecruitView is suitable because it contains interview-style responses and labels related to personality and soft-skill assessment. The dataset contains 2,011 naturalistic interview clips from more than 300 participants and 27,000 pairwise comparative judgments [3].

Since the goal is audio-only modeling, the video component is used only to extract the speech signal. Facial expressions, visual features, and text transcripts are not used in the main models.

C. Data Splitting

A user-independent split should be used. This means that clips from the same participant should not appear in both training and testing sets. This prevents identity leakage, where the model learns speaker-specific patterns instead of general interview-delivery patterns.

TABLE II
Suggested Dataset Split

Split	Percentage	Purpose
Training	70%	Model learning
Validation	15%	Hyperparameter tuning and early stopping
Test	15%	Final model evaluation

D. Acoustic Feature Extraction

The acoustic feature model uses handcrafted features that are meaningful for interview delivery.

These features are useful not only for prediction but also for generating understandable feedback.

TABLE III
ACOUSTIC FEATURES AND THEIR RELEVANCE

Feature	Relevance to Interview Coaching
Duration	Indicates answer length
Speaking rate	Helps detect too fast or too slow speech
Pause ratio	Shows silence proportion in the answer
Long pause count	Indicates hesitation or planning gaps
Mean pitch	Captures general voice tone
Pitch standard deviation	Captures pitch variation and expressiveness
Mean energy	Captures loudness and engagement
Energy standard deviation	Captures variation in delivery
MFCC statistics	Captures spectral speech characteristics

E. Speech Embedding Extraction

Each audio file was loaded as a mono signal and resampled to 16 kHz before being passed into HuBERT. The hidden-state outputs produced by HuBERT model were mean-pooled across the time dimension to obtain one fixed-length embedding vector for each interview response.

F. Model Architectures

Three models were implemented and compared in this project: an acoustic feature baseline, a HuBERT embedding model, and a fusion model.

1) Acoustic Feature Baseline: The acoustic feature baseline used handcrafted speech-delivery features as input. The input vector consisted of acoustic indicators such as duration, energy statistics, pause ratio, voiced ratio, pitch statistics, and speaking rate. These features were passed to a Random Forest Regressor with 300 estimators.

Although this model is not a deep learning model, it was included as an interpretable baseline. It learns a mapping from manually extracted acoustic features to the four interview-related target scores. The output of the regressor is a four-dimensional vector:

$$\hat{y} = [\hat{y}_{speak}, \hat{y}_{conf}, \hat{y}_{interview}, \hat{y}_{overall}]$$

where the four values correspond to speaking skills, confidence score, interview score, and overall performance.

2) HuBERT Embedding Model: The HuBERT embedding model used pretrained HuBERT speech representations as input [7]. First, each audio response was passed through the frozen HuBERT encoder. The HuBERT model produced a sequence of hidden-state vectors across time. These hidden states were mean-pooled over the temporal dimension to obtain one fixed-length embedding vector for each audio sample.

The extracted HuBERT embedding was then passed to a multi-layer perceptron (MLP) regression head. The regression head used the following structure:

$$\text{HuBERT Embedding} \rightarrow \text{Linear} \rightarrow \text{ReLU} \rightarrow \text{Dropout}$$

$$\rightarrow \text{Linear} \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow \text{Linear Output Layer}$$

The final output layer produced four continuous values corresponding to the target scores.

3) Fusion Model: The fusion model combined pretrained HuBERT embeddings with handcrafted acoustic features. It used a two-branch neural network architecture.

The first branch processed the HuBERT embedding:

$$\text{HuBERT Embedding} \rightarrow \text{Linear} \rightarrow \text{ReLU} \rightarrow \text{Dropout}$$

The second branch processed the handcrafted acoustic feature vector:

$$\text{Acoustic Features} \rightarrow \text{Linear} \rightarrow \text{ReLU} \rightarrow \text{Dropout}$$

The outputs of both branches were then concatenated into a single fused representation:

$$z = [z_{hubert}; z_{acoustic}]$$

The output layer again produced four continuous target predictions. This architecture was designed to test whether explicit acoustic features such as pause ratio, speaking rate, pitch variation, and energy could add complementary information to the HuBERT speech embeddings.

G. Agentic Feedback Layer Architecture

The agentic feedback layer was designed to convert the numerical model outputs into understandable interview coaching feedback.

The feedback layer consists of four role-based agents: a judge agent, a critic agent, a supporter agent, and a coach agent. Each agent has a specific responsibility in the feedback process. The judge agent provides an overall assessment of the candidate's response based on the predicted scores. The critic agent identifies weaknesses or areas that may need improvement, such as low confidence, frequent pauses, low energy, or poor speaking flow. The supporter agent highlights positive aspects of the response, such as acceptable speaking rate, stable delivery, or stronger predicted scores. The coach agent converts the combined observations into practical improvement suggestions. Table IV the individual function of the agents [4].

The outputs from the individual agents are then combined into a final feedback structure. The final feedback contains an overall summary, strengths, areas to improve, actionable suggestions, a next practice task, a safety note, and a short record of the agent discussion.

TABLE IV
FEEDBACK AGENT ROLES

Agent	Function
Judge Agent	Gives an overall evaluation
Critic Agent	Identifies weaknesses and improvement areas
Supporter Agent	Highlights strengths
Coach Agent	Combines all feedback into final actionable advice

VI. Experimental Setup

This section describes the implementation environment, data preparation, model training configuration, and evaluation procedure used in the experiments.

A. Implementation Environment

The project was implemented in Python using Jupyter Notebook for model training, evaluation, result comparison, and visualization. Separate Python scripts were used for dataset preparation steps such as audio extraction, metadata cleaning, user-independent split creation, and acoustic feature extraction. The experiments were run on a local laptop environment.

The implementation was carried out using the following main libraries:

- `pandas` and `NumPy` for data loading, preprocessing, and array operations,
- `librosa` for audio loading and acoustic feature extraction,
- `scikit-learn` for the Random Forest baseline, feature scaling, and regression metrics,
- `PyTorch` for neural network training,
- `transformers` for loading the pretrained HuBERT model,
- `matplotlib` for plotting comparison results,
- `joblib` and `json` for saving trained models and result files.

No end-to-end fine-tuning of the HuBERT model was performed due to computational limitations. Instead, HuBERT was used as a frozen feature extractor.

B. Dataset Splitting

The RecruitView metadata was first processed to ensure that each row contained a valid extracted audio path. The dataset was then split into training, validation, and test sets using the participant identifier `user_no`.

The final split contained 1404 training samples and 328 validation samples, with the remaining samples used as the test set. The training set was used for model fitting, the validation set for early stopping in neural network models, and the test set for final evaluation.

C. Preprocessing and Feature Preparation

The original interview videos were converted into WAV audio files. For the pretrained speech model, audio was loaded as a mono signal and resampled to 16 kHz, which is the expected input sampling rate for the HuBERT model. To keep embedding extraction feasible on the available hardware, audio samples were limited to a maximum duration of 60 seconds.

Acoustic features were extracted from each audio file and saved into `acoustic_features.csv`. The extracted features included duration, energy statistics, pause ratio, voiced ratio, pitch statistics, and speaking rate. These features were later merged with the train, validation, and test split files using the `audio_path` column.

For the HuBERT-based models, each audio file was passed through the frozen HuBERT encoder. The hidden states produced by HuBERT were mean-pooled over time to obtain one fixed-length embedding vector per audio sample. These embeddings were saved as compressed `.npz` files for the training, validation, and test sets to avoid repeated extraction.

D. Model Training Setup

Three models were trained and compared.

- **Acoustic Baseline:** used the handcrafted acoustic feature vector as input and trained a Random Forest Regressor with 300 estimators to predict the four target scores: speaking skills, confidence score, interview score, and overall performance.
- **HuBERT embedding model:** The HuBERT embeddings were used as input to a multi-layer perceptron regression head. The HuBERT encoder itself was frozen, and only the regression head was trained.
- **Fusion model:** This model used two input branches: one branch for the HuBERT embedding and one branch for the handcrafted acoustic features. The outputs of the two branches were combined and passed through dense layers to predict the target scores.

All the input features and target scores were standardized before training. Mean squared error was used as the training loss, and Adam was used as the optimizer. Early stopping was applied based on validation loss to reduce overfitting. After prediction, the target scores were transformed back to their original scale before calculating evaluation metrics.

E. Model Comparison and Evaluation

The models were evaluated on the test set using five regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Pearson correlation, Spearman correlation, and coefficient of determination (R^2). MAE and RMSE measured prediction error, while Pearson and Spearman correlations measured how well the predicted scores followed the actual score trends. The R^2 score measured the proportion of variance explained by the model.

F. Agent Input Generation

After model comparison, the best model was selected as the final prediction layer. The predicted scores from this model were combined with acoustic indicators to create structured input for the agentic feedback layer.

VII. Results and Discussion

A. Results

This section presents the experimental results obtained from the implemented audio-only models. Three models were compared: the acoustic feature baseline, the HuBERT embedding model, and the proposed fusion model. The models were evaluated on the user-independent test set using MAE, RMSE, Pearson correlation, Spearman correlation, and R^2.

TABLE V
AVERAGE MODEL COMPARISON ACROSS ALL TARGET SCORES.

Model	MAE	RMSE	Pear.	Spear.	R^2
Acoustic Baseline	0.7449	1.1009	0.3786	0.4341	0.1105
HuBERT Embedding	**0.7082**	**1.0745**	**0.4138**	**0.4650**	**0.1536**
Fusion Model	0.7096	1.0781	0.3932	0.4448	0.1486

Table V shows the average performance of each model across the four target variables: speaking skills, confidence score, interview score, and overall performance.

The HuBERT embedding model achieved the best overall performance among the three compared models. It obtained the lowest average MAE of 0.7082 and the lowest average RMSE of 1.0745. It also achieved the highest average Pearson correlation of 0.4138, Spearman correlation of 0.4650, and R^2 score of 0.1536.

The acoustic feature baseline produced the weakest average performance among the three models. Its average MAE was 0.7449 and its average Pearson correlation was 0.3786. The proposed fusion model performed better than the acoustic feature baseline in terms of MAE, RMSE, and R^2, but it did not outperform the HuBERT embedding model overall.

Figure 2 shows the average MAE comparison across the three models. Since lower MAE indicates lower prediction error, the HuBERT embedding model achieved the best result in this comparison.

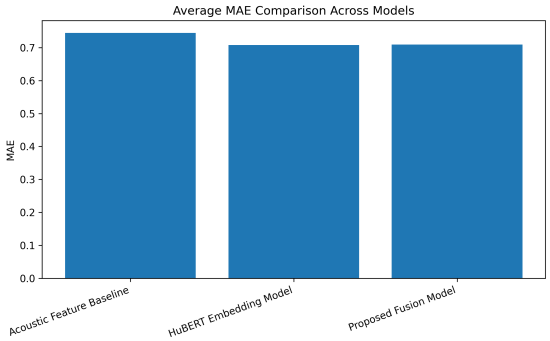


Fig. 2. Average MAE comparison across the acoustic baseline, HuBERT embedding model, and proposed fusion model.

Figure 3 shows the average Pearson correlation comparison across the three models. The HuBERT embedding model achieved the highest average Pearson correlation.

Table VI shows the per-target results of the HuBERT embedding model, which was the best-performing model overall.

TABLE VI
PER-TARGET PERFORMANCE OF THE HUBERT EMBEDDING MODEL.

Target	MAE	RMSE	Pear.	Spear.	R^2
Speaking Skills	0.6915	1.0040	0.3963	0.4751	0.1331
Confidence	0.6942	1.0676	0.4138	0.4430	0.1550
Interview Score	0.7284	1.1986	0.4137	0.4582	0.1533
Overall Perf.	0.7189	1.0278	0.4313	0.4837	0.1730

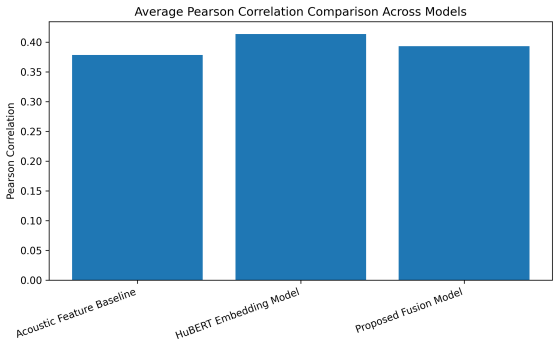


Fig. 3. Average Pearson correlation comparison across the acoustic baseline, HuBERT embedding model, and proposed fusion model.

Among the individual target scores, the HuBERT embedding model achieved the lowest MAE for speaking skills and confidence score. The highest Pearson correlation was observed for overall performance.

Prediction-versus-actual scatter plots were also generated for the proposed fusion model. Figure 4 shows the predicted and actual values for the overall performance target. The points are concentrated around the central score range, with fewer samples appearing at the extreme score values.

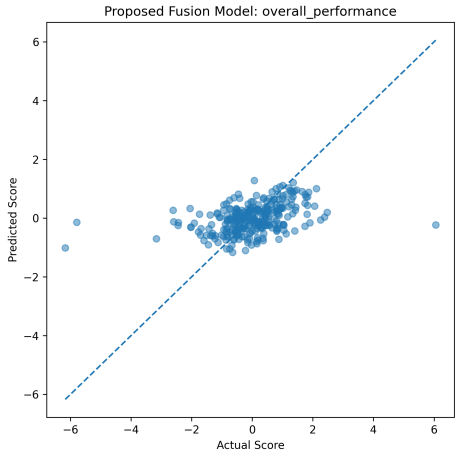


Fig. 4. Predicted versus actual values for the overall performance target using the proposed fusion model.

The final output of the deep learning layer was converted into a structured JSON format for the agentic feedback layer. For demonstration, selected test samples were passed to the feedback agents. Figure 5 shows an example of the generated feedback output for the question *"Tell me about yourself"* using the input file `demo_agent_input.json`.

B. Discussion

The experiments were guided by two hypotheses. H1 stated that combining pretrained HuBERT speech embeddings with handcrafted acoustic features would improve interview-score prediction compared with using HuBERT embeddings alone. The results did not support this hypothesis. The fusion model

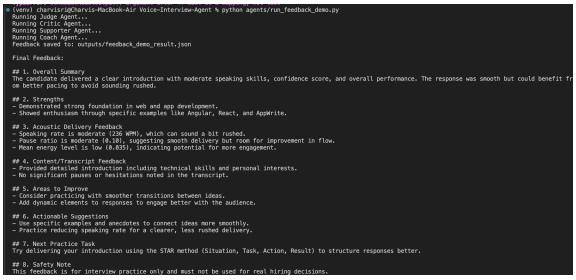


Fig. 5. Example feedback generated by the agentic feedback layer using predicted scores and acoustic indicators.

performed better than the acoustic feature baseline, but it did not outperform the HuBERT embedding-only model. The HuBERT embedding model achieved the best overall performance across the average metrics.

The HuBERT embedding model also outperformed the acoustic feature baseline across all average metrics. This indicates that pretrained speech representations captured more useful information from the audio signal than the manually extracted acoustic features alone. This supports the use of self-supervised speech representations for speech-based prediction tasks, since such models can learn useful acoustic and linguistic patterns from raw speech signals [6], [5], [8] .

The acoustic feature baseline still produced meaningful results. Although it had higher error than the HuBERT-based models, it achieved moderate correlation values. This shows that simple speech-delivery indicators such as pause ratio, speaking rate, pitch, and energy are still useful for understanding interview performance. Previous interview and behavioural analysis studies have also used prosodic and acoustic cues for predicting communication-related traits and interview performance [1], [9].

The fusion model was expected to improve prediction performance by combining two types of information: deep speech embeddings and explicit acoustic indicators. One possible reason is that HuBERT embeddings already encode many acoustic and prosodic patterns, such as speech rhythm, vocal variation, and speaking style [7]. Therefore, the handcrafted acoustic features may not have added enough new information to improve the prediction results. Another possible reason is that the fusion architecture used in this project was simple and not extensively optimized due to time and computational limitations.

H2 stated that a multi-agent feedback layer could make the system more useful for interview coaching by transforming numerical prediction scores and acoustic indicators into understandable and actionable feedback. The design of the role-based feedback layer is motivated by multi-agent evaluation frameworks, where agents with different roles can contribute different perspectives before producing a final output [4]. The final system therefore uses the HuBERT embedding model as the prediction layer and retains the acoustic features as interpretable indicators for the agentic feedback layer. This is important because numerical scores alone are not very useful for a user. For example, a predicted confidence score or speaking skills score can indicate performance level, but it does not explain what the user should improve. By combining predicted scores with acoustic indicators, the feedback layer can generate more understandable comments, such as pointing out frequent pauses, low vocal energy, fast speaking rate, or limited pitch variation.

Overall, the results show that deep speech embeddings are useful for audio-only interview score prediction, while acoustic features remain valuable for explanation and feedback generation. The system is not intended for automated hiring decisions, but for helping users reflect on their speaking style, confidence, and delivery.

VIII. Conclusion, Limitations, and Future Work

This project developed an audio-only deep learning based voice interview coach using the RecruitView dataset. The system used extracted interview audio to predict four interview-related scores: speaking skills, confidence score, interview score, and overall performance. Three models were implemented and compared: an acoustic feature baseline, a HuBERT embedding model, and a fusion model combining HuBERT embeddings with handcrafted acoustic features.

The results showed that the HuBERT embedding model achieved the best overall performance across the evaluation metrics. It performed better than the acoustic feature baseline and slightly better than the fusion model. This suggests that pretrained speech embeddings can capture useful information from interview audio for score prediction. The fusion model did not improve over the HuBERT-only model, but the acoustic features remained useful for the agentic feedback layer because they provide interpretable cues such as pause ratio, speaking rate, pitch variation, and energy.

The final system therefore uses deep learning predictions together with acoustic indicators to generate structured interview coaching feedback. Instead of only showing numerical scores, the agentic layer converts the predicted scores and acoustic indicators into human-readable feedback using judge, critic, supporter, and coach agents. This makes the system more useful as a coaching tool for improving speaking style, confidence, and interview delivery.

A. Limitations

There are several limitations in the current implementation. First, the transcript was not used as a model input, so the system does not deeply evaluate the semantic quality, relevance, or structure of the answer. This limits the prediction of targets that may depend strongly on content, such as answer quality or overall interview performance.

Second, the HuBERT model was used only as a frozen feature extractor. The full model was not fine-tuned on RecruitView due to time and computational constraints. Fine-tuning may improve performance but would require more computational resources and careful regularization.

Finally, the handcrafted acoustic features were extracted using predefined signal-processing rules. While useful, these features may not capture all relevant speaking behaviours.

B. Future Work

A possible future direction is to replace or extend the manually engineered acoustic feature extraction with a learned audio-behaviour representation module. Instead of relying only on predefined features such as pitch, energy, and pause ratio, a deep learning model could be trained to learn higher-level delivery patterns directly from speech, such as hesitation, fluency, vocal confidence, and speaking engagement. These learned indicators could then be provided to the feedback agents along with the predicted interview scores.

Another future improvement would be to extend and combined into a multimodal coaching system with agentic feedback. This can make model outputs more understandable for users. The system should be viewed as an interview coaching assistant rather than an automated hiring or candidate evaluation tool.

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